

BHANU NAGPURE

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SUMMARY

Flutter & Python engineer with 2+ years of hands-on experience building and shipping cross-platform mobile and web applications. Creator of PyMaster — a production Flutter app live in 170+ countries with a 5.0 Play Store rating. Proven track record through international research internships (Mitacs/Canada), 2 published papers (ACM, IGI Global), a granted patent, and multiple national hackathon wins. Seeking Flutter Developer or Full Stack roles where I can own features end-to-end and ship real products.

TECHNICAL SKILLS

Languages: Dart, Python, Kotlin, Java, JavaScript, HTML, CSS, SQL, C++

Mobile & Frontend: Flutter, React, Android SDK, Riverpod 2.0, Clean Architecture, REST APIs, Firebase

Backend & Cloud: Firebase (Firestore, Auth, Cloud Functions), Google Cloud, Node.js basics

AI / ML: TensorFlow, PyTorch, Scikit-learn, Transformers/BERT, OpenCV, Gemini LLM, on-device AI (LiteRT)

Tools & Practices: Git, GitHub, CI/CD basics, Agile, UI/UX Design, Postman, RevenueCat, Google Mobile Ads

PROFESSIONAL EXPERIENCE

Flutter Developer & Founder

Aug 2025 – Present

Devanshu Studios (PyMaster) · Remote / India

- Architected and shipped PyMaster — a production-grade, gamified Python learning platform on Google Play Store using Flutter and Clean Architecture (50k+ lines of code), available in 170+ countries with a 5.0 rating.
- Engineered a custom Native AI Tutor powered by the Gemini LLM for context-aware, in-app debugging assistance.
- Built an offline-first sync engine with batched Firebase writes to guarantee zero data loss during network interruptions.
- Implemented scalable state management using Riverpod 2.0 and monetization via RevenueCat (IAP) and Google Mobile Ads.

Machine Learning Intern

Aug 2024 – Oct 2024

Athabasca University · Athabasca, Alberta, Canada (Remote)

- Achieved 95% prediction accuracy on a text ideology identification model using Transformers and BERT.
- Built and deployed a prototype web interface enabling researchers to validate model outputs in real time.
- Presented research at the 3rd Annual TRESL Lab International Symposium, Athabasca, Canada.

Globalink Research Intern

Apr 2024 – Aug 2024

Mitacs · Edmonton, Alberta, Canada

- Led end-to-end development of MfieldTrip — a cross-platform (iOS, Android, Web) Flutter field-assignment app, owning UI/UX design, state management, and backend integration.
- Presented research findings at an international conference and authored comprehensive technical documentation.

Software Developer Intern

Oct 2022 – May 2024

BETiC, G H Raison College of Engineering · Nagpur, India

- Built an Android app in Kotlin for TMJ disorder screening, improving classification accuracy by 70% over the prior system.
- Contributed to a granted patent (No. 579268) for the Smart Chairside TMJ Examination Tool.
- Boosted camera diagnostic performance by 60% leveraging the Google ML Kit library.

Research Fellow

Mar 2022 – Oct 2022

Indian Institute of Technology Bombay · Mumbai, India

- Led a team of 5 to prototype a medical device for xerostomia patients, cutting diagnostic time by 80%.
- Won Most Ergonomic Device Award at MEDIC 2022 for the Xerostomia Estimator Device.

KEY PROJECTS

PyMaster — AI-Powered Python Learning App

Aug 2025 – Present

Flutter, Dart, Firebase, Riverpod 2.0, Gemini LLM, RevenueCat | [Play Store](#)

- Full production app: 3 learning tracks (Python Core, Data Science, Gen AI), gamified progress, offline-first sync, AI tutor, in-app purchases — built solo from scratch.
- 5.0 rating on Play Store, learners in 170+ countries.

Aegis-Edge — Offline Disaster Triage Android App

May 2026

Flutter, Android (CameraX), Gemma 4 (2B, on-device via LiteRT), RAG

- Built a 100% offline, multimodal (photo + voice + text) disaster triage app for the Google DeepMind Gemma 4 Good Hackathon.
- Implemented structured JSON output using START Triage Protocol with on-device Gemma 4 reasoning to eliminate hallucinations.

V-Safe-Anywhere — Women's Safety Wearable (IoT + AI)

Nov 2022 – Apr 2024

Flutter, Firebase, IoT, Google Apps Script

- Developed an IoT wearable with real-time Firebase sync, emergency SOS alerts, and 98% uptime.
- Published research paper in IGI Global (March 2024); secured Rs. 10 Lakh BIRAC E-YUVA Government Grant.

Autonomous Floating Garbage Collection Device

Jun 2023 – Sep 2023

Python, YOLOv5, NumPy, ESP32-CAM, Arduino

- Designed a computer vision system reducing manual waste collection by 60%; published in ACM ICACS 2024 (Hong Kong).
- Open-sourced on GitHub; presented at the 8th International Conference on Algorithms, Computing and Systems.

EDUCATION

B.Tech, Computer Science & Engineering

Dec 2021 – Jun 2025

G H Raison College of Engineering, Nagpur · CGPA: 8.7 / 10.0

PUBLICATIONS & PATENT

- V-Safe-Anywhere: Empowering Women's Safety with Wearable AI and IoT — IGI Global, March 2024
- Autonomous Floating Garbage Collection Device using Computer Vision and Robotics — ACM, ICACS 2024, February 2025
- Patent Granted: Smart Chairside TMJ Examination and Muscle Activity Detector Tool — Patent No. 579268 (Application 202321022476)

ACHIEVEMENTS

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| • Winner — SPIRIT '25 Health-Tech & AI Hackathon, IIT BHU Varanasi | Mar 2025 |
| • 1st Place — TechTrek Hackathon 2025, Nagpur | Jan 2025 |
| • 3rd Rank (State) — 16th Avishkar Inter-University Research Convention, MUHS Nashik | Jan 2024 |
| • Runner-up — SRISHTI 2023 National Level Project Competition, Kerala | Mar 2023 |
| • Runner-up — DIPEX 2023 State Level Exhibition, Amravati | Apr 2023 |
| • Rs. 10 Lakh Grant — BIRAC E-YUVA, Government of India | Feb 2023 |
| • Winner — MEDHA & MEDIC 2022, IIT Bombay (Most Ergonomic Device) | Oct 2022 |

OPEN SOURCE & CERTIFICATIONS

- Wikimedia Foundation — Scribe-Android contributor: implemented tablet keyboard layouts for 7+ languages (German, French, Spanish); fixed animation and theme bugs (Feb 2025 – Present)
- Certifications: Building Transformer-Based NLP Applications (Nvidia) · Postman API Fundamentals Student Expert · Medical Device Innovation Course (IIT Bombay)